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The Digital Dynamics of Language Vitality: Exploring Child-Algorithm Interactions on YouTube and Language Shift in Bishkek, Kyrgyzstan

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ABSTRACT

What is the role of children's interactions with YouTube's recommendation algorithm in language socialization, and could these interactions influence children's linguistic practices in such a way as to cause language shift? Our paper takes a mixed methodological approach to explore interactions between Kyrgyz families and YouTube, examining the reciprocal relationships between online content recommendations, offline linguistic behavior, and on/offline discourses. First, using ethnographic data, we investigate the belief that YouTube is causing language shift and the effect of this discourse on children's linguistic practices and language ideologies. Then we analyze results from an algorithmic audit of YouTube, demonstrating the proportion of Russian content presented to children. We argue that pressures from children's sociolinguistic environments cause them to prefer Russian, but children's preference for Russian is reinforced by material provided to them through YouTube's algorithm. Interactions with algorithms are language socializing and must be taken into account in situations of language shift.

Аннотация

Балдардын YouTube платформасынын сунуштоо алгоритми менен өз ара аракеттенүүсү тилдик социалдашуу процессинде кандай роль ойнойт жана мындай өз ара аракеттенүү балдардын тилдик практикаларына таасир этип, тилдин алмашуусуна алып келиши мүмкүнбү? Биздин макалада кыргыз үй-бүлөлөр менен YouTube'дун өз ара байланышын изилдөө үчүн аралаш методологиялык ыкма колдонулат: онлайн-контенттин сунушталышы, офлайн чөйрөдөгү тилдик жүрүм-турум жана онлайн/офлайн дискурстар ортосундагы өз ара байланыштар каралат. Биринчиден, этнографиялык маалыматтардын негизинде YouTube тил алмашууну шарттап жатат деген ишеним, ошондой эле бул дискурстун балдардын тилдик практикаларына жана тилдик идеологияларына тийгизген таасири изилденет. Андан соң YouTube'га жүргүзүлгөн алгоритмдик аудиттин жыйынтыктары талданып, балдарга сунушталган контенттин ичиндеги орус тилиндеги материалдардын үлүшү көрсөтүлөт. Биз балдардын орус тилин артык көрүшүнө алардын социолингвистикалык чөйрөсүндөгү басымдар себеп болорун, ошол эле учурда балдардын орус тилине болгон артыкчылыгы YouTube алгоритми аркылуу сунушталган материалдар менен дагы бекемделерин ырастайбыз. Алгоритмдер менен өз ара аракеттенүү тилдик социалдаштыруучу фактор болуп саналат жана тил алмашуу жүрүп жаткан шарттарда бул фактор сөзсүз түрдө эске алынышы керек.

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Аннотация

Какую роль взаимодействие детей с алгоритмом рекомендаций YouTube играет в языковой социализации и может ли это взаимодействие влиять на языковые практики детей таким образом, чтобы приводить к языковому сдвигу? В нашей статье используется смешанный методологический подход для изучения взаимодействия кыргызстанских семей и YouTube, а также взаимосвязи между рекомендациями онлайн-контента, языковым поведением в офлайн-среде и дискурсами в онлайн- и офлайн-пространстве. Во-первых, на основе этнографических данных мы исследуем распространённое убеждение о том, что YouTube способствует языковому сдвигу, а также влияние этого дискурса на языковые практики детей и их языковые идеологии. Затем мы анализируем результаты алгоритмического аудита YouTube, демонстрирующие долю русскоязычного контента, предлагаемого детям. Мы утверждаем, что давление со стороны социолингвистической среды детей способствует их предпочтению русского языка, однако это предпочтение дополнительно укрепляется материалами, которые предоставляет им алгоритм YouTube. Взаимодействие с алгоритмами является частью языковой социализации и должно учитываться при изучении ситуаций языкового сдвига.

Kids "[watch] these YouTube channels, cartoons, and bloggers. Parents work and give kids tablets, phones, computers, and it's all in Russian from their birth. And they start to speak Russian," Astra said, as she stole sips of lukewarm tea between offers to refill my cup, coaxing her four kids to eat, telling the youngest not to play with his food, and cutting off another argument over the Samsung tablet. I could hear the Europop-esque intro to the Russian-dubbed Korean cartoon *Super Wings* begin, as a little voice cried from the living room, "*GDE PULT'!?*" (Where's the remote!?) (excerpt adapted from fieldnotes 9/23/2022).

Like Astra, many adults in Bishkek, Kyrgyzstan, worry that children are growing up to speak only Russian and that their future grandchildren will no longer use Kyrgyz, their heritage language. They blame Russian-language media, particularly videos on YouTube, for their children's preference for Russian. Caregiver anxieties, however, counter most-used measurements of language vitality, which do not consider Kyrgyz to be threatened (Eberhard et al. 2021; UNESCO 2003).

Adults are not concerned that their children know Russian. Speaking Russian is still valued, and additive orientations to language learning prevail. Children are expected to grow up to be multilingual and speak Kyrgyz, Russian, and at least one additional "global" language (McDermott 2019). Caregivers worry that extensive exposure to Russian-language content online is driving preference for Russian to the exclusion of Kyrgyz. Adults fear children who do not speak Kyrgyz will not grow up to be good, truly "Kyrgyz" people with appropriate cultural values and competencies. Some caregivers try to "undo" this process (Kulick and Schieffelin 2004; Smith and Barad 2018) by controlling access to devices and the internet. Yet many adults see YouTube as a necessary childcare tool that engages children in situations where one adult may be responsible for multiple children and guests.

This paper examines how discourse surrounding YouTube's presumed role in Kyrgyz-to-Russian language shift influences children's language socialization in Bishkek. It also explores how children's interactions with YouTube's recommendation algorithm, within broader discursive and interactional contexts, have language-socializing effects that contribute to the shift. To explore how Kyrgyz families interact with YouTube's algorithm, we take a mixed methodological approach. YouTube's algorithm curates recommendations based on the search terms that children input and their viewing histories. Using ethnographic data from participant observation and recordings with Kyrgyz children and their families, we investigate children's linguistic practices and language ideologies in relation to the discourse that YouTube is the cause of language shift in Bishkek.

We then use an algorithmic audit of YouTube to determine how often Kyrgyz children are encountering Russian-language content. Using synthetic Kyrgyz user profiles based on data from participant observation, we determine the proportion of Russian content presented to children. We demonstrate that most content suggested to Kyrgyz children is in Russian, contrasting with the mixed-language recommendations made to Kyrgyz adults. Surprisingly, YouTube recommended fewer Kyrgyz-language videos to children who sought out and engaged with Kyrgyz content than to new users with no prior interaction history.

Russian-language content recommendations have consequences for how Kyrgyz children are socialized into and through language. We highlight the language-socializing effects of the interaction between children and YouTube's algorithm and explore the implications of algorithms on language ideologies and linguistic practices—in this case, in such a way as to precipitate shift. Humans are socialized into and through language (Ochs and Schieffelin 1984); repeated exposure to linguistic patterns in a particular cultural context creates an understanding of both how and when to use linguistic forms, who should speak in particular ways, and how to be a member of society. In the current study, children's preference for Russian is produced through repeated encounters across family interaction, peer talk, schooling, media, and algorithmically curated digital environments.

Foundational language-socialization research focused primarily on face-to-face interaction. This study builds on the literature that considers the role of multimodality in socializing interactions. Face-to-face verbal interaction happens concurrently with multiple "channels" of input, including gaze, body torque, and gesture, as well as interaction with "texts" and objects, which endlessly vary: from playing within the chalk squares of a hopscotch game to reading the words within a book (Goodwin 2000, 2007;

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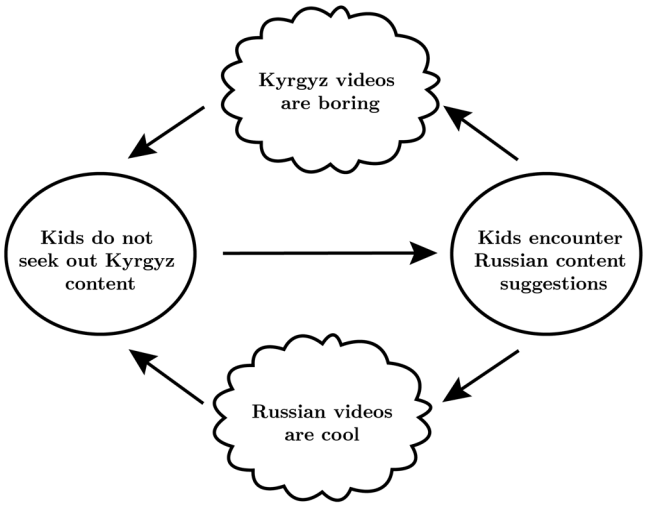


FIGURE 1 | Diagram of children's interactions with YouTube's recommendation algorithm. Diagram by Ashley McDermott, 2024.

Moore 2014). Socializing routines involve the coordination of talk, attention, bodies, spaces, and activities within a speech event. How interlocutors orient to each other and media/texts is socialized in culturally specific ways (see Heath 1983). Missing from work on multimodality and literacy in anthropology are accounts of how children are socialized through their interactions with online media, and more specifically, their interactions with algorithmic systems.

Rather than treating algorithms as text/media, we see algorithms as a highly constrained interlocutor and the platform as a channel of communication. Though YouTube's recommendation algorithm does not have intentionality in the same way as a caregiver or peer, it responds to children's actions and repeatedly organizes the linguistic environment available to them. Early work on language socialization treated children as novices socialized by adults; we draw on literature problematizing the "novice" and considering how children and peers are socializing agents (Garrett and Baquedano-López 2002; Goodwin and Kyratzis 2007; Namboodiripad and Henner 2023). Other research demonstrates how animals and even objects have a role in socializing routines (Berman and Smith 2021). We build on this expanded account of participation in language socialization to argue that interactions with algorithms are also language socializing.

Although engineers design algorithms for specific purposes, algorithms inadvertently reflect their creators' understandings of social phenomena (Seaver 2022) and react in real time to user input, oftentimes with unintended consequences. Viewers may be steered toward similar, highly viewed content, but no one receives uncurated access to the entirety of content available (Davidson et al. 2010; Wu et al. 2019). Each interaction with the platform further influences the available selections. In the case of Bishkek children's engagement with YouTube, these interactions with the algorithm form a feedback loop (Figure 1). Their perception of Russian content as appealing and Kyrgyz content as uninteresting influences their searches and viewing habits. Interactions with the algorithm amplify these preferences by primarily recommending Russian videos. Continued exposure associates Russian with entertaining content and the platform itself. Eventually this preference extends beyond YouTube.

To explore how children's interactions with algorithms are socializing, children's interactions have to be considered in the context of their overall socialization. We understand language socialization as occurring through various, concurrently open, "channels" in a network of language socialization (Smith and Barad 2018). Viewing socialization through networks consisting of a diverse "ecology of channels" (Kockelman 2010; Sherouse 2020, 7) including face-to-face communication with peers and families, schools, other internet media, and more, lets us explore how each is interrelated and exerts influence on each other. When a Bishkek kid makes a Russian-language search on YouTube, it is because they have been socialized into a preference for Russian through a variety of channels, not just YouTube alone. We argue that the culmination of socializing interactions in a Bishkek child's network of socialization causes them to prefer Russian, but children's preference for Russian and everyday interactions in Russian are reinforced by the material provided to them through YouTube, furthering language shift in the city.

Algorithms are integrated into daily life around the globe through online interactions. This paper emphasizes the impact of algorithms on language socialization, or how repeated *algorithmically curated* exposure to linguistic patterns in a particular cultural context creates an understanding of both how and when to use linguistic forms and how to be a member of society (Ochs and Schieffelin 1984). Language socialization is at the core of how humans form understandings of language and language users, form links between language and identity, and expand their linguistic repertoire. The language-socializing effects of algorithms are integral for understanding contemporary language use, and thus for understanding forms of language shift that may not be captured by conventional measures of vitality.

1 | Kyrgyz Historical and Ethnolinguistic Context

In northern Kyrgyzstan, Russian and Turkic language varieties have been in contact for approximately 130 years (Chotaeva 2004). In the 1920s, the USSR fostered the development of nations delimited by the state. Under the policy of *korenizatsiia*, or indigenization, the languages of titular nationalities were supported through language planning and policy. Later policies emphasized assimilation to Russian, which served as the language of interethnic communication, the language of higher education, the party, and government (Martin 2001; Smith 2006). Across the Soviet period, and especially in its later decades, local elites endorsed Russian to demonstrate political loyalty due to economic dependence on the center (Huskey 1995; Iğmen 2012).

During the Soviet era, most Bishkek residents were not predominantly Kyrgyz. Rather, post-Soviet migration led to demographic shifts (Table 1) (Alybaeva 2023; Chotaeva 2004). In the context of the current research conducted in 2022, participants stated that they perceived more Kyrgyz spoken on the street than in previous decades and attributed it to internal migration, though some posited that the children of internal migrants very quickly adapt to the Russian-dominant linguistic environment. This perception corresponds broadly to post-Soviet demographic shifts

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TABLE 1 | Russian and Kyrgyz demographic change after the fall of the USSR, 2022.

Nationality	1989 Census	2022 Census	Change (%)
Kyrgyz	2,229,663	4,995,877	124%
Russian	916,558	335,237	−63%

Note: Data from the National Statistics Committee of the Kyrgyz Republic. Table by the authors, 2024.

TABLE 2 | Kyrgyz respondents claiming Kyrgyz as their native language, 2009.

Nationality	Total L2 Speakers (*N*)	Kyrgyz	Russian	Uzbek	English	Turkish	German	Other
Kyrgyz	1,813,121	0.1%	94.9%	3.7%	0.5%	0.2%	0.0%	0.6%

Note: Data from the National Statistics Committee of the Kyrgyz Republic. Table by the authors, 2024.

TABLE 3 | L2 of Kyrgyz Respondents, 2009.

Nationality	Total Population	Kyrgyz as Native Language	Percentage (%)
Kyrgyz	3,804,788	3,799,385	99.9

Note: Data from the National Statistics Committee of the Kyrgyz Republic. Table by the authors, 2024.

(in Table 1), though census data do not directly capture language use; census-takers are reluctant to claim a "native language" that does not correspond to the conception of their own nationality (Table 2) (Shermatova et al. 2025). At the same time, Russian remained widely known in Bishkek: In 2009, 95 percent of Kyrgyz respondents reported proficiency in Russian as a second language (Table 3) .

The USSR's fall in 1991 prompted derussification policies reflecting heightened linguistic-ethnic national feeling (Pavlenko 2008). Early legislation sought to establish Kyrgyz as the state language, reflecting the national zeitgeist. Later policies alternated between strengthening Kyrgyz and adopting more moderate approaches. Though early legislation sought to make Kyrgyz the sole governmental language by 1998, this deadline was extended repeatedly until 2020 as moderate politicians in leadership sought to balance nation-building with protecting minority linguistic rights and appealing to internationalist factions (Huskey 1995; Liddicoat 2019; Orusbaev et al. 2008). Since the 1993 constitution, Kyrgyz has been the state language (**мамлекеттик тил**, государственный язык[1]) of the Kyrgyz Republic. Meanwhile, Russian was given official status by presidential decree in 1994 (**расмий тил**, официальный язык).

In contemporary language politics, those pursuing the "development"[2] of the Kyrgyz language draw on nationalist discourses equating Kyrgyz with Kyrgyz identity and fear of Kyrgyz loss (similar to Kazakhstan, cf. Smagulova 2021), but attempts at policy reform are tempered by Bishkek residents' actual linguistic practices and Kyrgyz proficiency, positive perceptions of the role of the Russian language, and officials' concerns over the republic's relationship with the Russian Federation. Within Kyrgyzstan, Russian media, including television channels, radio, social media, and print, is pervasive and seen as more trustworthy than both Kyrgyz and international outlets (IWPR 2023). The Russian Federation also exerts economic pressure to maintain pro-Russian language policy. For example, after a 2023 announcement by the chairman of the National Commission for the State Language and Language Policies regarding switching the script of Kyrgyz from Cyrillic to Latin, Russia suspended all dairy imports from Kyrgyzstan (see Marat and Kassymbekova 2023; Nourzhanov and Peyrouse 2021; RFE/RL 2023). Kyrgyzstan also remains economically dependent due to circular labor migration to Russia, with remittances comprising approximately 30 percent of the GDP (World Bank 2025). As of August 2025, President Sadyr Japarov, generally seen as a Kyrgyz nationalist, signed comprehensive legislation promoting Kyrgyz language use in public domains, though he has been critical of the policy and announced plans to expand Russian-language schooling (Pletneva 2025).

Instructional language in Kyrgyz schools reflects broader language policy trends. Kyrgyzstan has a multilingual educational system, including Kyrgyz medium of instruction (MOI), Russian MOI, mixed Kyrgyz/Russian MOI, and some Tajik and Uzbek MOI schools. In practice, school choice is restricted by region; most schools in rural Kyrgyzstan are Kyrgyz MOI, while as of 2012, 58.3% of mixed MOI and 71.1% of Russian MOI schools are located in or near Bishkek (Shamatov and Bahry 2020). Kyrgyz and Russian are both compulsory subjects from first grade until the end of secondary school, regardless of the school's MOI (Liddicoat 2019). In Bishkek, there are more Russian and mixed MOI schools than Kyrgyz schools (Bengard 2018). Still, crowding is a pervasive issue in Russian MOI classrooms, which generally have twice as many students as Kyrgyz MOI classrooms. Many Bishkek residents choose Russian-medium instruction because they associate Russian-language use with higher-quality education and better job opportunities; meanwhile, monolin-

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TABLE 4 | School Medium of Instruction in the city of Bishkek, 2025.

Medium of Instruction	Number of Schools
Kyrgyz	14 (13.6%)
Russian	16 (15.5%)
Bilingual	73 (70.9%)

Note: Data from Shermatova, Totobayev, and Bahry 2025; percentages calculated by the authors, 2024.

gual Kyrgyz speakers are stereotyped as backward, rural, and uncultured (McDermott 2026; Shermatova et al. 2025).

Given the complicated relationships with the Russian Federation and memories of the USSR, within Bishkek, it is controversial to term the Soviet Union "colonial."[3] While colonial consciousness and Indigeneity have been part of the public discourse and politics in Kazakhstan for some time (see Kudaibergenova 2016), in Kyrgyzstan debates over colonialism were becoming increasingly mainstream in 2022 and 2023 (see Kravtsova 2022). In participant observation, caretakers rarely discussed their understanding of Soviet influence in Kyrgyzstan as colonizing and largely viewed the USSR in a positive, nostalgic light (see Umetbaeva 2015). In contrast, many young adults interviewed encountered decolonial discourse as part of Kyrgyz language activism online. They saw an increase in this content after the start of the Russo-Ukrainian War and made connections between decolonial Ukrainian influencers and those in Central Asia (see also Alybaeva 2023; for similar discourse in Kazakhstan, see Kamalova 2025).

2 | Socializing Interactions With YouTube and Discourses of Shift

In Kyrgyzstan's contemporary linguistic environment, the online cannot be separated from the offline. By using both ethnographic methods and an algorithmic audit of YouTube, we gain a better understanding of the cumulative effect of multiple channels of communication on Bishkek children's language socialization. The qualitative analysis examines discourse about YouTube's russifying effect on children and on linguistic practices in everyday family interactions. The section "Ethnographically Informed Algorithmic Auditing" complements this analysis, demonstrating the connections between offline behavior and online recommendations.

Qualitative analysis draws on participant observation, semi structured interviews, and recordings collected over several months in 2016, 2018, and eight months between 2022 and 2023. In the main research phase (2022–2023), McDermott worked with 15 Kyrgyz families from various microdistricts, comprising 46 children aged nine months to 16 years. Families were recruited through network sampling. Most adults had moved to Bishkek as children, and almost all their children were born in the city. Middle- and high-income families were slightly overrepresented, but data from surveys and schools improved socioeconomic representation. Over 100 hours of audio and video recordings were collected.

McDermott conducted 30 interviews with adults, focusing on sociolinguistic backgrounds, language practices, and beliefs. An open-ended online survey was also distributed via Instagram, with over 100 responses. To examine discourses of language shift online, we systematically collected articles from Kyrgyz news sources shared on social media, along with the top 100 comments. Children's and parents' media interactions were also observed and recorded in a spreadsheet, noting both content type and language.

McDermott observed and recorded in three Bishkek schools: one private school (Jal Microdistrict) and two public schools (Vostok 5, Uchkun Zhilmassiv). Four families had children in private schools while the remainder attended public schools. Five children received Kyrgyz-medium instruction, but most studied in Russian. Despite these differences, all children used YouTube and nearly all preferred speaking Russian, even those in Kyrgyz-medium classes.

Language socialization is always mediated by language ideologies, or the "morally and politically loaded" "cultural systems of ideas about social and linguistic relationships" (Irvine 1989, 255). In Bishkek, caregiver concerns about children's Russian-language YouTube use are shaped by ideologies that link Kyrgyz language competence to Kyrgyz personhood and ethnonational continuity. Due to these ideologies, adults hear children's code-mixing, phonology, or hesitation in Kyrgyz as evidence of failed socialization into Kyrgyzness. Work on other shift contexts demonstrates that adult concerns about purity can produce shame, embarrassment, and avoidance between young speakers of minoritized languages (Bonner 2001; Garrett 2005; Messing 2007; Paugh 2014; Reynolds 2009).

The algorithmic audit and ethnographic data suggest that Bishkek children's preference for Russian could be linked to YouTube consumption and that this is a result of linguistic ideological assemblages discouraging code-mixing and ways of speaking other than an idealized, phonologically "Kyrgyz" Kyrgyz, as well as social YouTube viewing and peer use of Russian. Adults' claims that YouTube is making children Russian-speaking are themselves language-ideological accounts shaped by anxieties about screen time, Soviet and post-Soviet ethnonational categories, and shifting urban language hierarchies. Yet the algorithmic audit shows that these concerns are valid and that YouTube's recommender system does disproportionately expose child users in Bishkek to Russian-language content. The following sections explore both sides of this process: how adults interpret and respond to children's YouTube use and how children's interactions with YouTube become part of their broader socialization into Russian-dominant linguistic practices.

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3 | Screen Time and Development

Like parents worldwide, families in Bishkek encounter global discourses that screens are bad for their children. Negative views of screen time circulate in online parenting discussions and news, which cite scientific studies. When adults in Bishkek families blame YouTube for their children's preference for Russian, they draw on context that posits that screens harm their children's development.

While over the course of interviews children and parents reported that they spend at least two to three hours per day engaged with online media, due to stigma surrounding screen time, it is likely this number is much higher (see Livingstone and Blum-Ross 2020). Smartphones are ubiquitous in Bishkek due to low-cost entry-level models from China and South Korea. Most third spaces and many public spaces offer free wifi, while the fastest unlimited data plans are relatively inexpensive—10 dollars a month. Schools also use personal devices to address textbook shortages: Teachers ask kids to share pictures of textbooks and assignments via WhatsApp with students who do not have access to physical copies. Children spend much of the day watching videos on YouTube and TikTok, playing games, browsing social media, creating their own content, and learning via electronic channels.

In participant observation, adults consumed news almost entirely online through Instagram, and popular news accounts frequently featured articles regarding parenting and children's screen use. As Pravda.kg reports on February 14, 2024:

> Увеличение экранного времени у детей в двухлетнем возрасте приводит к нарушениям общения, мелкой моторики, личных и социальных навыков. К четырём годам также отмечается задержка развития в области общения и решении проблем. (Increased screen time in children as young as two years of age leads to impairments in communication, fine motor skills, and personal and social skills. By age four, there are also developmental delays in communication and problem solving.) (Loginov 2024)

Similar language is found in the Kyrgyz government's decision to ban TikTok, which the Ministry of Culture (**Маданият, маалымат, спорт жана жаштар саясаты министрлиги** 2023) posted to their Facebook page on August 30, 2023:

> TikTok **платформасын колдонууда балдардын ден-соолугуна жана өнүгүүсүнө зыян келтирүүчү маалыматты чектөөлөр жок жана ошондой эле курактык категорияны контролдоо функциялары киргизилбей жаткандыгы аныкталган**.
>
> (It was determined that there are no restrictions on information that harms children's health and development when using the "TikTok" platform, and also age category control functions are not included.)

Bishkek caregivers constantly consume media that tells them screen use is harming their children's development. Caregiver anxieties manifested mainly in attempts to control access to technological channels of communication (Smith and Barad 2018). When devices were not in the hands of children, adults hid phones, remotes, tablets, and laptops on top of refrigerators, bookshelves, and wardrobes. One mother controlled internet use by paying the weekly internet bill a day late, forcing her children to have a device-free day. Other caregivers tried to impose screen time limitations, such as only 30 minutes a day after finishing homework. The following sections explore how discourse surrounding screen time coalesces with purist language ideologies to create the specific fear that devices are causing children to become Russian.

4 | Language and Kyrgyz Identity

Caregivers in the study worried that children's screen use harmed both their overall development and their linguistic development and development into Kyrgyz people. Adults were concerned that their children were losing their identity as *Kyrgyz*. They equate the use of the Kyrgyz language with Kyrgyzness, and speaking Russian with acting like Russians (McDermott 2019). In post-Soviet contexts, ideologies of nationality have often intensified associations between titular languages and titular ethnic identities, even when everyday practice remains multilingual and mixed (Pavlenko 2008; Smagulova 2021). These associations are sociohistorically produced ways of interpreting linguistic difference and assigning social value to speakers (Irvine 1989; Irvine and Gal 2000; Kroskrity 2018, 2021).

> Throughout participant observation, people directly equated language to behavior:
>
> "My parents' generation says that today's young children are spoiled [*izbalovanie*]. They get what they want. They speak like Russians; they behave like Russians." (Interview 7/2018)

Others cited the lack of knowledge about specific practices, like the distribution of horsemeat at community gatherings, as examples of Russian-speaking Kyrgyz children becoming less Kyrgyz.

Links between language and identity, and the fear of loss, may fuel concerns about linguistic purism. Online, popular Instagram bloggers post nearly daily memes regarding "correct" use of Kyrgyz and discouraging viewers from using Russian lexical items in their Kyrgyz speech. For example, in the following post from the Instagram account Tilimpoz, with over 264K followers, they invite viewers to "**Кел** чисто **кыргызча сүйлөшөбүз**" (Come, we speak pure Kyrgyz!).

In the post (Figure 2), the black text, "чисто" (pure), is Russian, while the rest of the text (gray) is in Kyrgyz. Tilimpoz promotes

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FIGURE 2 | Post from September 7, 2023, with over 5000 likes. Instagram/Tilimpoz (screenshot by Ashley McDermott), 2023.

what they see as "pure" Kyrgyz and mocks the code-mixed way most Bishkek adults speak.

Adults were especially attuned to their children's code-mixing. In the following fieldnote excerpt, Astra and her four children are gathered for breakfast when five-year-old Umai attempts to use a Russian verb in a structurally Kyrgyz utterance:

> Umai wanted the remote, wanted to play, but also wanted to finish eating her cereal. She then said, "**Men suu** khochu" (I want water). Astra corrected her with "**Men suu ichem**" (I drink water). (Fieldnotes 10/6/2022)

Astra suggests "**ichem**" (I drink) in Kyrgyz instead of the Russian verb "khochu." She changes the meaning of what Umai intends from "I want water" to "I drink water." "I want water" would require the dative case in Kyrgyz. The correction demonstrates her negative orientation toward language mixing and is an implicit assessment of her daughters' competency.

A few weeks later, Astra's 11-year-old daughter Aksaamai similarly corrected an utterance of Umai's when they were watching YouTube in the living room just before dinner. Umai said, "Ya khochu **tamak**" (I want food), and Aksaamai corrected her with "Ya khochu edu." Aksaamai demonstrates that she too has an idea of "correct" speech that does not allow for mixing, but she prefers Russian.

Adults frequently corrected and commented on the phonological features of their children's Russian and Kyrgyz speech. For example, here Astra corrects how her eight-year-old son says the name of the lake "Issyk Kol," the largest lake in Kyrgyzstan:

> Astra talked about how long it took to get to Almaty, comparing it to Issyk Kul but not Naryn. Aidil' said "**Issyk Kul**," and his mom corrected him to the Kyrgyz "**ө**." He asked what "**kol**" meant; she clarified it meant "lake," and he said it meant "goryachoe ozero" (hot lake, Russian). (Fieldnotes 3/27/2023)

She repeats the vowel "**ө**" (œ) after he says "y" (u). "**ө**" represents a phoneme that does not occur in standardized Russian.

Caregivers believed their children spoke Kyrgyz incorrectly and "with an accent" due to perceived code-mixing and phonological differences:

1. Ну они наверное понимают. Даже если используют, ну я думаю что **мындай таза эмес да**, с акцентом. Я так думаю.
2. Well, they probably understand [Kyrgyz]. Even if they use it, well, I think that it's not clean; it's with an accent. I think this. (Interview 12/2022)
3. У нее тоже пятеро же детей. Старшая сестра. Они тоже кыргызский **класска барышат**. Все. Но они в основном разговаривают на русском. Они хотя с первого класса ходят на кыргызский. Ну с первого класса кыргызский язык обучения у них, наоборот акцент на кыргызском у них есть. Они дома только на русском разговаривают.
4. She also has five children. Older sister. They also go to Kyrgyz class. All of them. But they basically speak in Russian. Although from the first class they went to Kyrgyz [school]. Well, from the first class they had Kyrgyz as the language of instruction; conversely, they have an accent in Kyrgyz. They only speak Russian at home. (Interview 12/2022)
5. **Эжеке урушат экен эме депчи "кыргызча сүйлөшкүлө" деп урушат экен. Эжекенин көзүнчө кыргызча аракет кылышат экен, бирок** все равно **орусча сүйлөшөт**. **Биздин балдарыбыз** ещё **кыргызчаны жакшы билишет**, а там **мондай балдар бар: Баягы кыргыз тилди аябай акцент менен сүйлөгөн**, и **аз сүйлөгөн балдар бар**.
6. Older sister [auntie] scolds, saying, "Speak in Kyrgyz." They try to speak Kyrgyz in front of their aunt, but they still speak Russian. Our children still understand Kyrgyz well, but there are children like this: There are children who speak the Kyrgyz language with a strong accent, and there are children who speak it less. (Interview 12/2022)
7. И дети у меня почти что на кыргызском не умеют говорить, то есть они понимают, когда я что-то говорю, но отвечают они мне на русском. Вот это плохо. Мне кажется это проблема номер один, это их надо приучать и у них какой-то акцент, как будто это не их родной язык.
8. And my children hardly know how to speak Kyrgyz; that is, they understand when I say something, but they answer me in Russian. This is bad. It seems to me that this is problem number one; they need to be taught, and they have some kind of accent, as if it is not their native language. (Interview 1/2023)

In the above examples, four different mothers relate that children in their family speak Kyrgyz with an "accent." The first states that the language is not "clean"; it is incorrect because it is not "pure" Kyrgyz. The third interviewee says that her kids' Auntie "scolds" them when they don't speak Kyrgyz to her, implying that they are being disrespectful or disobedient. The fourth states outright that she believes it is bad that her children "hardly know" Kyrgyz.

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She relates their speech to knowing Kyrgyz so poorly that it is "as if it is not their native language." The parents suggest that their children and the children around them speak Kyrgyz improperly and disapprovingly equate the speech with that of foreigners.

Caregivers' understanding of Russian-language content online and their children's linguistic practices is tied to their beliefs regarding the implications of screen time for child development and presumed connections between language and identity stemming from Soviet and post-Soviet discourse surrounding nationality. Many language-socializing routines adults enact due to these fears of Russian-language screen time creating less "Kyrgyz" behaviors in their children, such as correcting mixed speech, have the unintentional effect of discouraging children from further speaking Kyrgyz. As in many other cases of language shift, children use and know language in ways that adults do not always expect, and do so according to their own language ideologies formed by exposure to different patterns of use (Garrett 2005; Kulick 1992; Meek 2007; Meek 2019; Ochs and Schieffelin 1984; Paugh 2014).

5 | YouTube and Russian

Throughout participant observation, interviews, and surveys, adults attributed their children's dispreference for and "incorrect" use of Kyrgyz to YouTube. As anthropological work on algorithms and media demonstrates, the accounts people relate regarding platforms influence user reactions to them (see Gershon 2010; Ruckenstein 2023). In Bishkek, this often manifests as the story of a relative's child in their natal village who spontaneously begins speaking Russian due to YouTube. Such stories reflect adult beliefs that YouTube has a massive and pernicious influence on children's linguistic practices: Even in spaces perceived to be free of Russian-language influence, children are speaking Russian.

In Bishkek, most parents give their children access to devices. The content the children encounter on the devices, particularly on YouTube, is in Russian. Adults believe that because of the exposure to content in Russian, children speak Russian. Across surveys and interviews, caregivers echoed this sentiment:

1. Дети говорят на русском, потому что они смотрят на русские мультфильмы, они пользуются гаджетами, где тоже не всё переведено, не весь интерфейс имеет кыргызский текст, поэтому у них в практике, получается, активно становится русский язык, да.
2. Children speak Russian because they watch Russian cartoons; they use gadgets where not everything is translated, [the gadget's interface] isn't all in Kyrgyz text, so while they are using it, it's all Russian in practice. (Interview 12/2022)
3. Дети сейчас очень часто сидят в телефонах и там тоже я не слышу что на кыргызском что-то было там в основном на русском.
4. Children very often sit on the phone, and there I don't hear anything in Kyrgyz; basically, everything that is there is in Russian. (Interview 1/2023)
5. Дети в основном смотрят тик-ток, ютуб, они на русском, на русском больше наверное. Со мной, например, **кыргызча ырларды ырдашат, орусча ырларды ырдашат**. В основном у них русский, и в садике на русском разговаривают.
6. Children mostly watch TikTok and YouTube; they are in Russian, probably mostly in Russian. With me, for example, they sing Kyrgyz songs and Russian songs. Mostly they speak Russian, and they speak Russian in kindergarten. (Interview 1/2023)

In the examples from different caregivers, watching videos is either explicitly given as a reason for their children's use of Russian (1), or it is implied as they move directly from speaking about their children's device use to their linguistic practices (3).

6 | Social Viewing

Consistent with caregiver concerns, children may acquire certain forms of Russian and are partly socialized into a preference for Russian through YouTube. Prior studies indicate that without copresent interaction, children cannot learn a language from video or will only learn some lexical items (DeLoache et al. 2010; Kuhl et al. 2003; Rice et al. 1990). However, L2 research suggests that there are contexts in which unintentional language acquisition from screen time is possible (Simon et al. 2016; Sundqvist 2009).

The type of exposure that matters for both acquisition and socialization depends on cultural and interactional context (Casillas et al. 2021; Rogoff 2003; Rowe and Weisleder 2020). Research in linguistic anthropology emphasizes repetition within routines, where linguistic and social practices predictably co-occur (Moore 2014). Research in language acquisition suggests that a relationship exists between frequency of input and acquisition, but how much input is necessary is unclear (Kidd and Donnelly 2020). Infrequent interactions can be socializing, and what children attend to and find significant may determine how and what they learn (Robinson 1995; Tomasello 2010).

The interactions Bishkek children have with algorithmically curated content through YouTube occur within certain contextual conditions which can lead to language learning from videos (see Roseberry et al. 2014). First, children often watch YouTube in the context of offline peer social interactions, where YouTube is but one channel of communication occurring concurrently with many others. Second, YouTube content is central to children's offline social lives, and semiotic resources from online content are decontextualized and recontextualized offline.

As kids socialize together, YouTube is but one phatic channel in a given moment and is not always the one children are specifically attending to (Figure 3).

1. Nurlan: "Кукареку, кукареку. Кукареку." (Cock-a-doodle-doo, cock-a-doodle-doo. Cock-a-doodle-doo.)
2. Emir: "Пожалуйста отдай пожалуйста пульт." (Please, please give [me] the remote.)
3. Bakyt: "Я не знаю где стоит пульт." (I don't know where the remote is.)

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FIGURE 3 | Eleven children "watch" YouTube in a child's bedroom at a family dinner party. Screenshot from audiovisual data captured and edited by Ashley McDermott, 2023. [This figure appears in color in the online issue]

4. Nurlan: "Кукарекуку, кукарекуку, кукареку, кукареку, кукареку, кукареку, кукареку." (Cock-a-doodle-doo [repeats six more times].)
5. Bakyt: "Это же маленький пульт, маленький пульт." (It is such a small remote, a small remote.)
6. Emir: "Эй. Будьте тише. Блин где этот пульт?" (Hey. Be quiet. Dang, where is that remote?)
7. Nurlan: "Кукареку, (кукареку. Кукареку, кукареку. Кукареку. Кукареку, кукареку.)" (Cock-a-doodle-doo [repeats six times].)
8. Umar: "Давайте это, давай это, давай это. Это, это Поззи, Поззи, Поззи." (Let's do this, let's do this, let's do this. This, this is Pozzi, Pozzi, Pozzi.)
9. Nurlan: "Кукареку, кукареку." (Cock-a-doodle-doo, cock-a-doodle-doo.)
10. YouTube: "Привет, меня зовут Поззи. Попробуй не засмеяться челлендж." (Hello, my name is Pozzi. Try the don't-laugh challenge.)
11. Nurlan: "Это слайм, слайм." (This is slime, slime.)

In this example, 11 children are gathered in a child's bedroom during a dinner with extended family. Children direct others to selectively attend to YouTube videos in an environment where kids are talking, playing with a chicken-shaped slingshot, running around, and playing games on their phones. Joint attention is constructed through positive stances toward YouTube and specific types of content, which children come to align with (Bucholtz 2009; Du Bois 2007; Goodwin and Kyratzis 2007). First, 15-year-old Bakyt and 10-year-old Emir bicker about the remote. Emir is focused on finding it and shushes his five-year-old cousin Nurlan, who repeatedly says "cock-a-doodle-doo" throughout the interaction (lines 1, 4, 7, 9). The remote and the ability to control the YouTube content are constructed as important. Once Emir finds the remote, the other kids take notice and weigh in on the selection of content. Eight-year-old Umar is excited about a thumbnail of Russian video game streamer and influencer Pozzi and repeats himself several times, convincing everyone to watch (line 8). Nurlan abandons his slingshot and crowing to watch the slime in the video (line 11). This short interaction demonstrates how, as children watch together, they are also socialized into how to relate to certain types of content, what to pay attention to, how to search, and how to utilize "YouTube" as a channel of communication. The search function returns results primarily in Russian regardless of the search term; content that kids find interesting is provided in Russian, and these offline face-to-face interactions regarding YouTube are also in Russian.

The mass-mediated effects of YouTube can be seen in how children bring semiotic resources from videos into their offline daily lives. Despite the illusion that an individual's algorithmic profile provides tailored content, the algorithm provides content proven to keep viewers watching. Ultimately, this leads to more suggestions from channels that are already highly viewed (Wu et al. 2019). Children often use features of Russian speech that they are exposed to through YouTube, such as lexical items and phrases associated with their favorite videos. For example, though most children in Bishkek did not pay for streaming services, Wednesday Addams was wildly popular. Over the course of participant observation in April 2023, children repeatedly sang and quoted a fan-made song called "Девочка Уэнсдей" (Girl Wednesday) from a YouTube video published in March 2023 with over 8,000,000 views. Kids adopted fashion similar to that of Wednesday and had accessories with pictures of the character. Similarly, children knew about games such as *Minecraft* and *Poppy Playtime* without experiencing the media firsthand. Kids learned about Wednesday and their other interests, such as

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anime, sports, or video games, through both their peers and YouTube. Citational acts of such content are essential in the performance of youth identities and the mediation of community (Nakassis 2016; Spitulnik 1996).

Children form preferences for trendy content largely in offline, face-to-face interactions with peers. They are more likely to initiate a search for this kind of content, but they are also more likely to be recommended the content. Kids learn Russian and learn to prefer Russian from YouTube because the context of viewing is social and because Russian associations with mass-mediated content and children's varied on/offline interests permeate other domains of use (a type of semiotic "bundling"; Keane 2003).

7 | Ethnographically Informed Algorithmic Auditing

We use ethnographically informed algorithmic auditing to produce a quantifiable picture of the proliferation of Russian online, an idealized picture of the content environment a Kyrgyz child would experience on YouTube in Bishkek. We treat algorithmic auditing, however, with the same critical lens as our discourse analysis; the content that children are exposed to on YouTube is algorithmically curated by recommender systems that respond to a child's input. The terms children search, the video thumbnails they select, and their watch histories are informed by language-socializing interactions through various on/offline channels in networks of socialization.

Algorithmic auditing tests black-box systems like YouTube, whose internal workings are not directly inspectable by external researchers. The method originated in mid-20th-century audit studies by social scientists and activists, which aimed to quantify discrimination in housing and employment. To determine whether an employer used discriminatory hiring practices, a researcher could submit comparable resumes that varied only by applicant gender and observe whether men received higher response rates. Algorithmic audits vary the inputs to a black-box system and record its outputs. Though this approach does not reconstruct internal processes, it can be used to detect patterns in algorithmic behavior. Algorithmic audits should not be understood as revealing immutable characteristics of algorithms, but rather reflecting the operation of a system during the audited time period. In fact, several audits have prompted system designers to update their algorithms, with follow-up audits finding reductions in problematic behavior (Metaxa et al. 2021).

We conduct a sock-puppet algorithmic audit to investigate the content YouTube's recommender systems prioritize for child users in Kyrgyzstan. In a sock-puppet audit, computer programs emulate user activity by carrying out interactions with platforms (Escher and Banovic 2020; Sandvig et al. 2014). We instructed our bots to interact with the YouTube platform and programmatically record the video recommendations on each page. We used three types of sock-puppet bots to investigate how the platform responds to child and adult users, running all bots in separate application instances on open access points in Bishkek. We encoded instructions for the bots in Python, managed web browser interaction with Selenium WebDriver, and stored data in SQLite.

The first type of bot establishes a baseline for the proportion of languages in YouTube recommendations in Kyrgyzstan. It opens the YouTube homepage and records the top six recommended videos. We do not set any demographic information or signal user interests, and the bot captures the initial set of videos recommended to any user, child or adult.

The second type of bot determines the proportion of languages returned for child- and adult-interest searches. It inputs a term into the search bar, presses enter, and records the returned video recommendations. To identify terms, we combined ethnographic observations of family search histories with nationally representative data from Google Trends. Google Trends provides the most popular searches in Kyrgyzstan for the "Children's Interest" category but does not provide a basis for its categorization. To confirm that the category constructed by Google matched what we knew about the behavior of Kyrgyz children, we compiled a list of search terms from participant observation and search histories from participant devices and compared the data to Google Trends. We corroborated 35 of 74 searches, and the remaining terms appear plausible; for example, we did not observe "винни пух" (Winnie the Pooh), but the term is an instance of the observed genre "мультики" (cartoons).

The third type of bot determines the proportion of languages encountered by child users browsing the platform. Each sock-puppet bot watches a series of 10 Kyrgyz-language child-interest videos to signal user demographics to the platform. Many YouTube analytics features are unavailable for Kyrgyzstan, so we could not directly query the platform for the videos most popular with child users. Instead, we use the most popular results encountered by our search bots. This technique leverages cookie-based personalization: YouTube can track users across a browsing session and update the videos that it recommends by inferring users' interests based on the content users consume. After the bot watches the videos, it performs eight browsing steps (i.e., clicking on a video recommendation, autoplaying a recommended video, or navigating back to the front page). Then, it records the top six recommended videos.

8 | Content Suggestions for Kyrgyz Kids

We ran our auditing study in Bishkek during summer and fall 2023. Our experiments identified a total of 1653 unique videos. Kyrgyz university students watched each video and hand-labeled languages written or spoken.

First, we analyzed the video recommendations given to Kyrgyz users prior to any user personalization. Our first type of bot visited the YouTube homepage (Figure 4) and recorded the top six recommendations in 232 independent trials (Figure 5). We determined how many Russian, Kyrgyz, and English videos appeared for each trial.

We conclude that Russian- and Kyrgyz-language videos are recommended at similar rates when the YouTube algorithm lacks information to personalize recommendations based on predicted user demographics.

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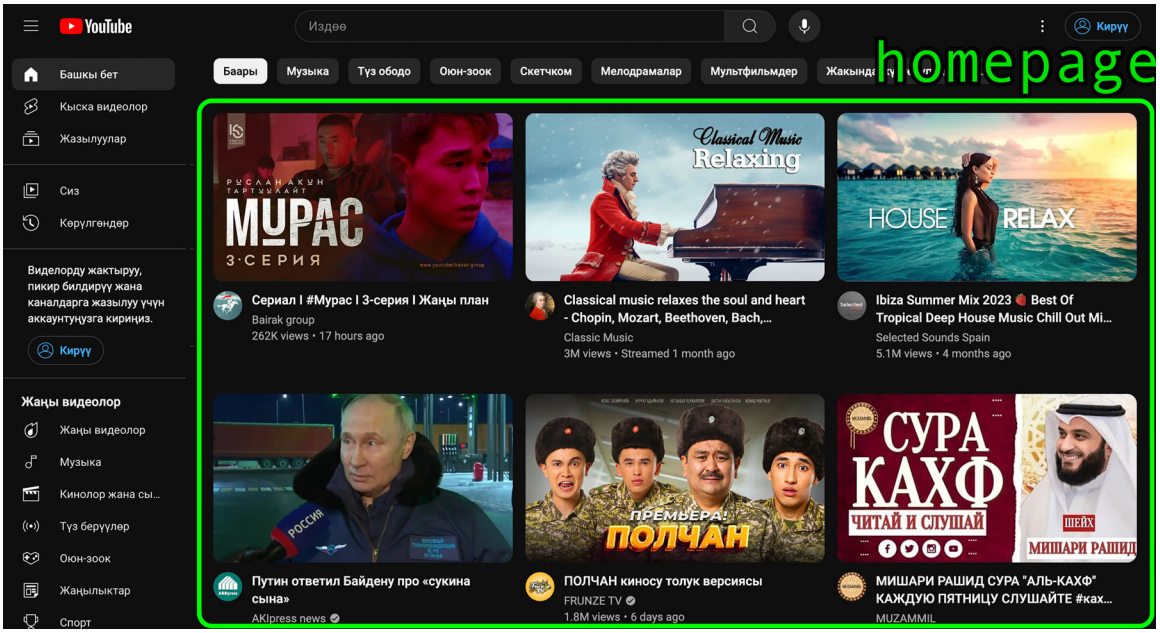


FIGURE 4 | Screenshot of the YouTube homepage. This image has two Kyrgyz-, two Russian-, and two English-language video recommendations in the top six homepage recommendations. YouTube (screenshot by Nel Escher), 2024. [This figure appears in color in the online issue]

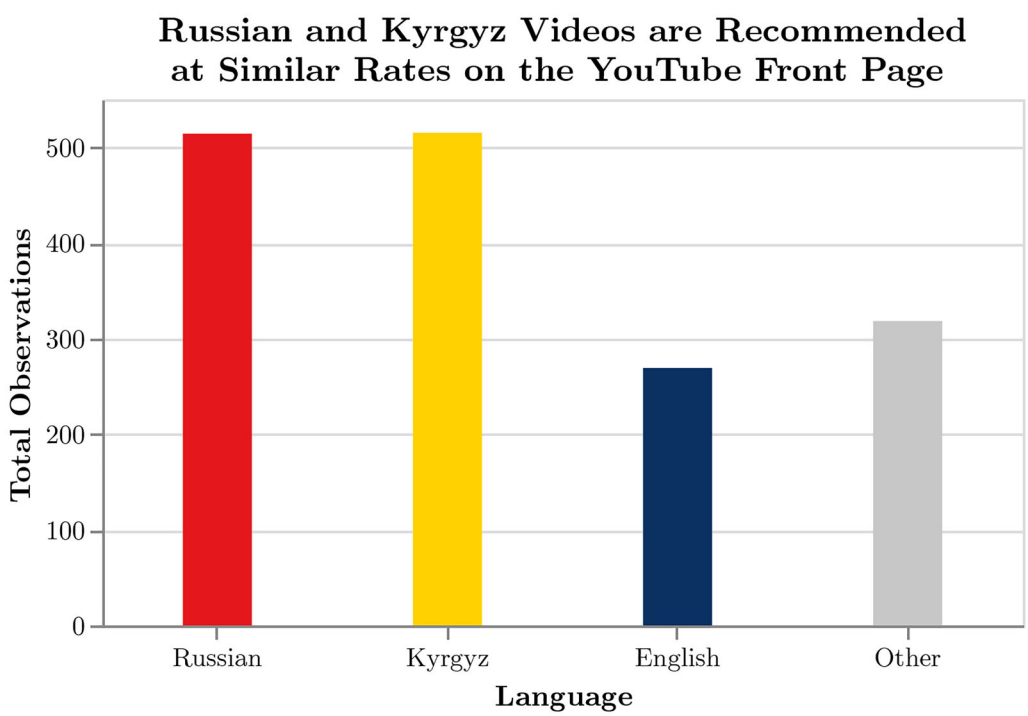


FIGURE 5 | Bar chart showing how often Kyrgyz, Russian, English, and other language results appeared in the top six video recommendations on the YouTube homepage across 232 trials. Russian- and Kyrgyz-language videos appeared in the top six recommendations at the same average frequency ($M_{Kyrgyz} = 2.2$, $M_{Russian} = 2.2$), which was higher than English-language videos ($M_{English} = 1.2$). Chart by Nel Escher, 2024. [This figure appears in color in the online issue]

Next, we analyzed the recommendations when searching for adult and child-interest terms. Our second type of bot searched 85 adult- and 60 child-interest terms from Google Trends (omitting 16 overlapping terms) and recorded the top 10 recommendations returned for each search (Figure 6).

Our analysis suggests that YouTube prioritizes Russian-language videos over Kyrgyz-language videos for popular children's search terms. The word cloud visualization in Figure 7 demonstrates how the linguistic diversity of Bishkek, on/offline, is simplified, as overwhelmingly Russian results are returned in response

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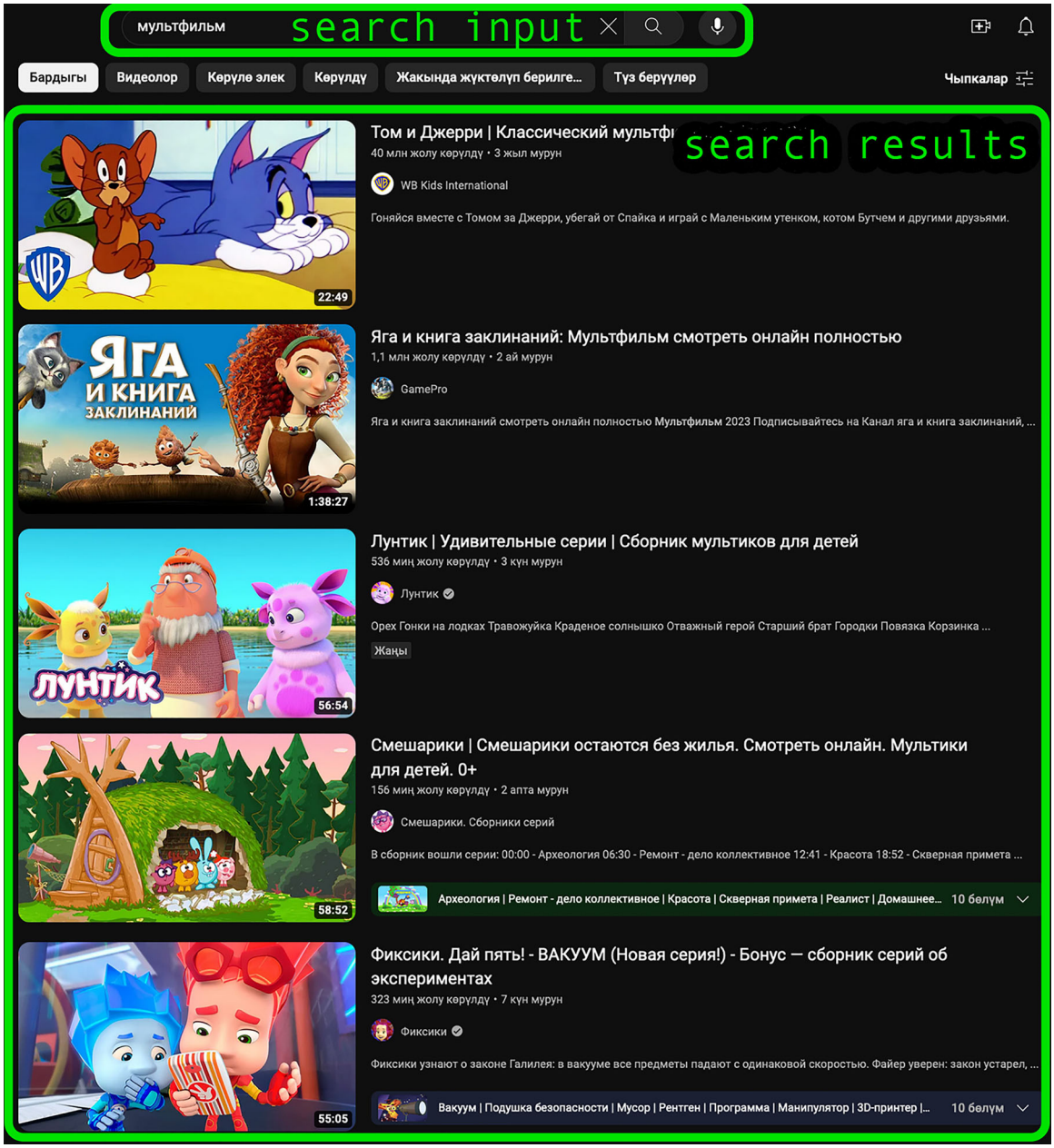


FIGURE 6 | Screenshot of the search page. In this image, the term "мультфильм" (cartoon), which is used in both Kyrgyz and Russian, returns only Russian results. YouTube (screenshot by Nel Escher), 2024. [This figure appears in color in the online issue]

to children's searches as opposed to the adult searches in Figure 8.

Lastly, we analyzed the recommendations for child users with Kyrgyz-language preferences in Bishkek. Our third type of bot watched a series of videos to personalize the recommendations, performed eight navigation steps, and recorded the top six recommended videos in 84 independent trials.

The YouTube algorithm does not prioritize Kyrgyz videos for children who signal a strong interest in Kyrgyz-language content. Even after watching 10 videos in Kyrgyz, simulating an artificially strong preference, our child persona bots were recommended fewer Kyrgyz-language videos than the nonpersonalized bots ($M_{KyrgyzChild} = 1.1$, $M_{KyrgyzNonpersonalized} = 2.2$; see Figure 9). This difference is not due to a dearth of Kyrgyz-language child-interest videos, as we identified 287 unique Kyrgyz-language child-interest videos throughout these experiments. YouTube recommends fewer Kyrgyz-language videos to children who have sought out and engaged with Kyrgyz content than to users who do not yet have viewing histories.

Children select content differently than bots; they are more inclined to choose videos from recommendation thumbnails than from a collection of bookmarked URLs. Thus, these results depict a scenario that is disproportionately advantageous for the generation of Kyrgyz-language recommendations aimed at children's interests. The findings reflect an idealized environment rather than a realistic one.

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FIGURE 7 | Word cloud of video titles returned for adult-interest searches. Terms that appear in Russian-language videos are colored in red, Kyrgyz in yellow, English in blue, and other or uncertain terms in gray, following the manual labels. If a term occurs across videos in different languages, we color it with the language most frequently represented in those videos. Word cloud by Nel Escher, 2024. [This figure appears in color in the online issue]



FIGURE 8 | Word cloud of video titles returned for child-interest searches. Russian terms are colored in red, Kyrgyz in yellow, English in blue, and other or uncertain terms in gray. Word cloud by Nel Escher, 2024. [This figure appears in color in the online issue]

9 | Discussion: Russian-Speaking Kyrgyz Kids and YouTube

YouTube's ranking and recommender systems filter billions of videos into a tractable set of suggestions for a user. Driven to satisfy unarticulated desires, YouTube uses recent site activity and internal scoring systems to decide which videos to promote (Davidson et al. 2010). Online platforms like YouTube optimize their algorithms to draw users to their services and hold their attention (Seaver 2022). Although recommender systems can match user interests, curating content to boost user engagement can have unintended downstream effects.

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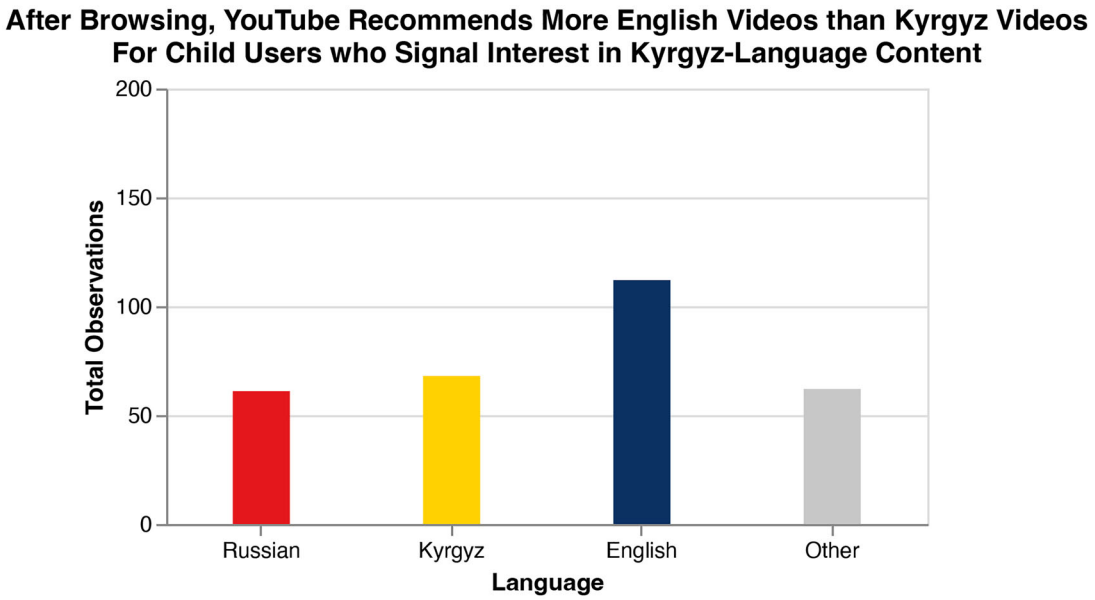


FIGURE 9 | Bar chart showing how often Kyrgyz, Russian, English, and other language results appeared in the top six video recommendations after bots signaled interest in *Kyrgyz-language child* content, then browsed for eight steps across 84 trials. English-language videos appeared in the top six recommendations at a higher frequency ($M_{English} = 1.8$) than Kyrgyz or Russian results ($M_{Kyrgyz} = 1.1$, $M_{Russian} = 1.0$). Chart by Nel Escher, 2024. [This figure appears in color in the online issue]

Optimizing algorithms to ensnare user attention leads to the proliferation of certain strains of content, while crowding out other categories. Most views are concentrated in a small subset of videos on YouTube, with recommendations shuttling users between content that is already popular on the platform (Wu et al. 2019). This concentration of attention shapes how groups are portrayed and made visible (Noble 2018). Some feel erased by recommender systems, noting that creators with marginalized social identities appear less frequently in platform suggestions (Karizat et al. 2021). Other forms of content may be magnified; media reports have charged YouTube with pushing young men into "rabbit holes" of far-right politics, offering increasingly extreme content suggestions that can lead to radicalization (Abbruzzese 2019).

Algorithmic categorizations do not necessarily match users' offline identities and are generally not known to them (Cheney-Lippold 2017; Ruckenstein 2023; Seaver 2022). Despite mismatches, algorithms influence and limit expressions of identity. Our algorithmic audit documents the proliferation of Russian-language content on YouTube. Though no evidence exists that Silicon Valley software developers intend this behavior, the recommendation algorithm repeatedly exposes Kyrgyz children to Russian-language content. Even if children attempt to make their Kyrgyz-speaking identity legible to the algorithm, they are rebuffed (see Colbjørnsen 2018; Gaw 2022). Repeated exposure shifts children's preferences to align with their algorithmically assigned identities.

User preference for Russian cannot fully explain this outcome. Languages are not equally amplifiable; no rabbit hole is formed for Kyrgyz-language content if children repeatedly seek it out. While rabbit holes have been studied as a means of political radicalization, we show that platforms can also steer users into linguistic rabbit holes—environments where only a privileged subset of languages can proliferate.

Socialization through YouTube is possible due to the culmination of the entire network of socialization, rather than one channel. First, purist language ideologies, equating and conflating language with identity, and stigmatization of screens lead parents to fear for their children's development as Kyrgyz people. As a result, code-mixing is less tolerated. When children are corrected and hear that they speak "with an accent," they are paradoxically dissuaded from speaking Kyrgyz and begin to strongly prefer Russian. Research on language contact and shift links children's embarrassment and ambivalence toward their heritage language to direct and indirect experiences of shame and adult discourses of linguistic purity (e.g., Bonner 2001; Garrett 2005; Kroskrity 2018; Messing 2007; Paugh 2014; Reynolds 2009). Children's Kyrgyz is evaluated by adults as accented or mixed, while Russian is repeatedly associated with trendy content and positive peer experiences. YouTube does not create this hierarchy alone, but it amplifies it by making Russian-language content more frequently available. While adults code-mix frequently and prefer phonologically "Kyrgyz" forms of lexical items, children mix less, prefer to speak Russian in most domains, and often use more "Russian" phonology when speaking Kyrgyz.

Second, watching YouTube in Bishkek is a social activity. Children collaboratively search for, watch, and discuss videos. They talk about videos with their peers and deploy the linguistic and semiotic resources from videos in their offline lives. Joint attention and engagement might be sufficient for language learning, as they are socialized into a preference for Russian-language content.

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10 | Conclusion

Despite a wealth of multilingual and Kyrgyz-language content on YouTube, children in Bishkek are influenced by the almost exclusively Russian content recommended by YouTube's algorithm. YouTube is not simply a neutral repository of available Kyrgyz, Russian, and English content. It is an interactional and algorithmic environment that filters and ranks some linguistic resources over others.

The Russian-language content provided through algorithmic interactions is integral to children's social life. Repeated exposure to certain types of Russian-language content helps shape children's preferences for that content. Meanwhile, through shared viewing and discussion, Russian-language media becomes ingrained in children's everyday interactions. Thus, children's interactions with algorithms are language socializing; they shape linguistic practices, language ideologies, and the way children perceive themselves and others in their communities. The semiotic resources children have access to are in part determined by the content suggested by algorithms, yet also determined by children's interpretations, choices, and habits as they interact with the platform.

Caregivers' concerns about children's Russian-language media consumption are tied to fears that Russian-speaking children will not be properly socialized into, or develop into, ethnolinguistically Kyrgyz people. They hear children's code-mixing, Russian phonology, and reluctance to speak Kyrgyz as signs of a broader loss of Kyrgyz cultural knowledge, values, and norms. Yet the corrective practices adults use to address this perceived loss may have the opposite effect: When adults criticize mixed speech and accented Kyrgyz, the language is associated with negative evaluations of competence both in the language and as an ethnically Kyrgyz person.

Though adult worries stem from ideological assemblages, algorithms do play a role in how children are socialized in and through language. In this case, the culmination of these interactions is contributing toward a shift, notwithstanding the official status of the Kyrgyz language as "institutional," "vital," and thus supposedly not under threat (Eberhard et al. 2021; UNESCO 2003). Frameworks such as UNESCO's and Ethnologue's emphasize factors such as absolute number of speakers, intergenerational transmission, official status, use in education, institutional support, literacy materials, and presence in public or media domains. By these measures, Kyrgyz appears comparatively secure. Ninety-nine point nine percent of ethnic Kyrgyz census respondents report speaking Kyrgyz; Kyrgyz is the state language; and it is supported institutionally and compulsory in schools (National Statistics Committee of the Kyrgyz Republic 2022).

NGO labels portray a very different picture of Kyrgyz use in Kyrgyzstan from that described by Bishkek caregivers. These measures do not consider the algorithmically curated linguistic environments in which children increasingly spend their time. A language may be institutionally supported, taught in schools, and widely claimed in census categories while still being marginalized in digital spaces. To understand language vitality, algorithmic interactions must be taken into account—otherwise, as in the case with Bishkek families, shifts may be occurring in ways that are unaccounted for in current measures. Throughout the world, algorithms are deeply interwoven with digital communication and interaction and need to be considered a fundamental aspect of language socialization and language use broadly.

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Endnotes

[1] All Kyrgyz text (both Cyrillic and transliterated) is bolded throughout to distinguish it from Russian.

[2] Press coverage of government language policy and policies themselves frame language revitalization as "development" of the language or "raising the level" of use and knowledge of the language rather than "language endangerment."

[3] The scholarship on the USSR maintains a divided orientation toward Soviet colonialism, with one side emphasizing the state's modernizing agenda and nationalities policy, and the other highlighting associations between Soviet policies and a "civilizing mission," assimilationist policies, and the economic exploitation of the periphery (for a more thorough review, see Sultanalieva 2023).

References

Abbruzzese, J. 2019. "New Zealand Shooting Aftermath, Tech's Role in Spreading Extremism Comes Under Scrutiny." NBC News, March 19, 2019.

Alybaeva, A. 2023. "Decolonial Discourse in Kyrgyzstan: Language as a Tool of National Identity Building." MA thesis, American University of Central Asia.

Bengard, A. 2018. "Only 216 Schools in Kyrgyzstan use Russian as Language of Instruction." 24.Kg, February 27, 2018. https://24.kg/english/77135/.

Berman, E., and B. Smith. 2021. "De-Naturalizing the Novice: A Critique of the Theory of Language Socialization." *American Anthropologist* 123, no. 3: 590–602.

Du Bois, J. W. 2007. "The Stance Triangle." In *Stancetaking in Discourse: Subjectivity, Evaluation, Interaction*, 139–182. John Benjamins.

Bonner, D. 2001. "Garifuna Children's Language Shame: Ethnic Stereotypes, National Affiliation, and Transnational Immigration as Factors in Language Choice in Southern Belize." *Language in Society* 30, no. 1: 81–96.

Bucholtz, M. 2009. "From Stance to Style: Gender, Interaction, and Indexicality in Mexican Immigrant Youth Slang." In *Stance: Sociolinguistic Perspectives*, edited by Alexandra Jaffe, 146–170. Oxford University Press.

Casillas, M., P. Brown, and S. C. Levinson. 2021. "Early Language Experience in a Papuan Community." *Journal of Child Language* 48, no. 4: 792–814.

15481433, 0, Downloaded from https://anthrosource.onlinelibrary.wiley.com/doi/10.1111/aman.70136 by Univ of California Lawrence Berkeley National Lab, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Cheney-Lippold, J. 2017. We are Data: Algorithms and the Making of our Digital Selves. New York University Press.

Chotaeva, C. 2004. "Ethnicity, Language, and Religion in Kyrgyzstan." PhD diss, Tohoku University.

Colbjørnsen, T. 2018. "My Algorithm: User Perceptions of Algorithmic Recommendations in Cultural Contexts." In *Human-Machine Communication: Rethinking Communication, Technology, and Ourselves*, edited by A. Guzman, 167–183. Peter Lang.

Davidson, J., B. Liebald, J. Liu, P. Nandy, and T. Van Vleet. 2010. "The YouTube Video Recommendation System." In Proceedings of the Fourth ACM Conference on Recommender Systems, 293–296. ACM.

DeLoache, J., C. Chiong, and K. O'Doherty. 2010. "Do Babies Learn From Baby Media?" *Psychological Science* 21, no. 11: 1570–1574.

Eberhard, D. M., G. F. Simons, and C. D. Fennig. 2021. *Ethnologue: Languages of the World*. 24th ed. SIL International.

Escher, N., and N. Banovic. 2020. "Exposing Error in Poverty Management Technology: A Method for Auditing Government Benefits Screening Tools." *Proceedings of the ACM on Human-Computer Interaction* 4, no. CSCW1: 1–20.

Garrett, P. B. 2005. "What a Language Is Good for: Language Socialization, Language Shift, and the Persistence of Code-Specific Genres in St. Lucia." *Language in Society* 34, no. 3: 327–361.

Garrett, P. B., and P. Baquedano-López. 2002. "Language Socialization: Reproduction and Continuity, Transformation and Change." *Annual Review of Anthropology* 31: 339–361.

Gaw, F. 2022. "Algorithmic Logics and the Construction of Cultural Taste of the Netflix Recommender System." *Media, Culture and Society* 44, no. 4: 706–725.

Gershon, I. 2010. "Breaking up Is Hard to do: Media Switching and Media Ideologies." *Journal of Linguistic Anthropology* 20, no. 2: 389–405.

Goodwin, C. 2000. "Action and Embodiment Within Situated Human Interaction." *Journal of Pragmatics* 32, no. 10: 1489–1522.

Goodwin, C. 2007. "Participation, Stance, and Affect in the Organization of Activities." *Discourse and Society* 18, no. 1: 53–73.

Goodwin, M. H., and A. Kyratzis. 2007. "Children Socializing Children: Practices for Negotiating the Social Order Among Peers." *Research on Language and Social Interaction* 40, no. 4: 279–289.

Heath, S. B. 1983. *Ways With Words: Language, Life, and Work in Communities and Classrooms*. Cambridge University Press.

Huskey, E. 1995. "The Politics of Language in Kyrgyzstan." *Nationalities Papers* 23, no. 3: 549–572.

Iğmen, A. 2012. *Speaking Soviet With an Accent: Culture and Power in Kyrgyzstan*. University of Pittsburgh Press.

Irvine, J. 1989. "When Talk Isn't Cheap: Language and Political Economy." *American Ethnologist* 16, no. 2: 248–267.

Irvine, J. T., and S. Gal. 2000. "Language Ideology and Linguistic Differentiation." In *Regimes of Language: Ideologies, Polities, and Identities*, edited by Paul Kroskrity, 35–83. School of American Research Press.

IWPR. 2023. Narratives and Perceptions of Russian Propaganda in Kyrgyzstan. Institute for War and Peace Reporting and EUCAM.

Kamalova, A. 2025. "Speak Kazakh: Language Ideologies in Kazakhstan's Social Media in Times of the Russian–Ukrainian War." *Journal of Sociolinguistics* 29: 182–193.

Karizat, N., D. Delmonaco, M. Eslami, and N. Andalibi. 2021. "Algorithmic Folk Theories and Identity: How TikTok Users Co-Produce Knowledge of Identity and Engage in Algorithmic Resistance." *Proceedings of the ACM on Human-Computer Interaction* 5, no. CSCW2: 1–44.

Keane, W. 2003. "Semiotics and the Social Analysis of Material Things." *Language and Communication* 23, no. 3–4: 409–425.

Kidd, E., and S. Donnelly. 2020. "Individual Differences in First Language Acquisition." *Annual Review of Linguistics* 6: 319–340.

Kockelman, P. 2010. "Enemies, Parasites, and Noise: How to Take up Residence in a System Without Becoming a Term in It." *Journal of Linguistic Anthropology* 20, no. 2: 406–421.

Kravtsova, V. 2022. "Feminism, Nationalism, Decolonization: Perspectives From Bishkek and Almaty." *Berliner Blätter* 85: 75–86.

Kroskrity, P. V. 2018. "On Recognizing Persistence in the Indigenous Language Ideologies of Multilingualism in two Native American Communities." *Language and Communication* 62: 133–144.

Kroskrity, P. V. 2021. "Language Ideological Assemblages Within Linguistic Anthropology." In *Crossing Borders, Making Connections: Interdisciplinarity in Linguistics*, edited by Allison Burkette and Tamara Warhol, 129–165. Mouton de Gruyter.

Kudaibergenova, D. T. 2016. "The use and Abuse of Postcolonial Discourses in Post-Independent Kazakhstan." *Europe-Asia Studies* 68, no. 5: 917–935.

Kuhl, P. K., F.-M. Tsao, and H.-M. Liu. 2003. "Foreign-Language Experience in Infancy: Effects of Short-Term Exposure and Social Interaction on Phonetic Learning." *Proceedings of the National Academy of Sciences* 100, no. 15: 9096–9101.

Kulick, D. 1992. *Language Shift and Cultural Reproduction: Socialization, Self, and Syncretism in a Papua new Guinean Village*. Cambridge University Press.

Kulick, D., and B. B. Schieffelin. 2004. "Language Socialization." In *A Companion to Linguistic Anthropology*, edited by Alessandro Duranti, 349–368. Wiley.

Liddicoat, A. J. 2019. "Language-in-Education Policy in the Central Asian Republics of Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan." In *The Routledge International Handbook of Language Education Policy in Asia*, edited by Andy Kirkpatrick and A. J. Liddicoat, 452–470. Routledge.

Livingstone, S., and A. Blum-Ross. 2020. *Parenting for a Digital Future: How Hopes and Fears About Technology Shape Children's Lives*. Oxford University Press.

Loginov, O. 2024. "Gadgets and Kids: How Screens Affect a Child's Development." Pravda February 14, 2024.

Marat, E., and B. Kassymbekova. 2023. "Reclaiming the Narrative: Decolonising Central Asian Studies for a More Inclusive Understanding." *APSA Comparative Politics Newsletter* 33, no. 1: 9–12.

Martin, T. 2001. The Affirmative Action Empire: Nations and Nationalism in the Soviet Union, 1923–1939. Cornell University Press.

McDermott, A. E. 2019. "The Linguistic Landscape of Post-Soviet Bishkek." *Region: Regional Studies of Russia, Eastern Europe, and Central Asia* 8, no. 2: 227–255.

McDermott, A. E. 2026. "Raising Kyrgyz Kids: Family Language Policy, Childrearing, and Language Shift in a Bishkek Family." *International Journal of Bilingual Education and Bilingualism* 29, no. 8: 1–21.

Meek, B. A. 2007. "Respecting the Language of Elders: Ideological Shift and Linguistic Discontinuity in a Northern Athapascan Community." *Journal of Linguistic Anthropology* 17, no. 1: 23–43.

Meek, B. A. 2019. "Language Endangerment in Childhood." *Annual Review of Anthropology* 48, no. 1: 95–115.

Messing, J. 2007. "Multiple Ideologies and Competing Discourses: Language Shift in Tlaxcala, Mexico." *Language in Society* 36: 555–577.

Metaxa, D., J. S. Park, R. E. Robertson, et al. 2021. "Auditing Algorithms: Understanding Algorithmic Systems From the Outside In." *Foundations and Trends in Human–Computer Interaction* 14, no. 4: 272–344.

Moore, L. C. 2014. "Language Socialization and Repetition." In *The Handbook of Language Socialization*, edited by Alessandro Duranti, Elinor Ochs, and Bambi B. Schieffelin, 169–189. Wiley.

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Nakassis, C. V. 2016. *Doing Style: Youth and Mass Mediation in South India*. University of Chicago Press.

Namboodiripad, S., and J. Henner. 2023. "Rejecting Competence: Essentialist Constructs Reproduce Ableism and White Supremacy in Linguistic Theory." *Language Learning* 73, no. S2: 321–324.

National Statistics Committee of the Kyrgyz Republic. 2022. Population and Housing Census of the Kyrgyz Republic for 2022.

Noble, S. U. 2018. *Algorithms of Oppression: How Search Engines Reinforce Racism*. New York University Press.

Nourzhanov, K., and S. Peyrouse. 2021. *Soft Power in Central Asia: The Politics of Influence and Seduction*. Lexington Books.

Ochs, E., and B. Schieffelin. 1984. "Language Acquisition and Socialization: Three Developmental Stories." In *Culture Theory: Essays on Mind, Self, and Emotion*, edited by R. A. Shweder and R. A. LeVine, 276–320. Cambridge University Press.

Orusbaev, A., A. Mustajoki, and E. Protassova. 2008. "Multilingualism, Russian Language, and Education in Kyrgyzstan." *International Journal of Bilingual Education and Bilingualism* 11, no. 3–4: 476–500.

Paugh, A. 2014. *Playing With Languages: Children and Change in a Caribbean Village*. Berghahn Books.

Pavlenko, A. 2008. "Multilingualism in Post-Soviet Countries: Language Revival, Language Removal, and Sociolinguistic Theory." *International Journal of Bilingual Education and Bilingualism* 11, no. 3–4: 275–314.

Pletneva, S. 2025. "Kyrgyz Authorities Have Tightened Requirements for the use of the National Language." Forbes Russia, August 11, 2025. https://www.forbes.ru/society/543699-vlasti-kirgizii-uzestocili-trebovania-po-ispol-zovaniu-nacional-nogo-azyka.

Reynolds, J. 2009. "Shaming the Shift Generation: Intersecting Ideologies of Family and Linguistic Revitalization in Guatemala." In *Native American Language Ideologies: Beliefs, Practices, and Struggles in Indian Country*, edited by Paul V. Kroskrity and Margaret C. Field, 213–237. University of Arizona Press.

RFE/RL's Kyrgyz Service. 2023. "Russia Suspends Dairy Products From Kyrgyzstan After Calls in Bishkek to Drop Cyrillic Script." Radio Free Europe/Radio Liberty, April 21, 2023. https://www.rferl.org/a/russia-suspends-kyrgyzstan-dairy-products-cyrillic-script/32371757.html.

Rice, M. L., A. C. Huston, R. Truglio, and J. Wright. 1990. "Words From 'Sesame Street': Learning Vocabulary While Viewing." *Developmental Psychology* 26, no. 3: 421–428.

Robinson, P. 1995. "Attention, Memory, and the 'Noticing' Hypothesis." *Language Learning* 45, no. 2: 283–331.

Rogoff, B. 2003. *The Cultural Nature of Human Development*. Oxford University Press.

Roseberry, S., K. Hirsh-Pasek, and R. M. Golinkoff. 2014. "Skype Me! Socially Contingent Interactions Help Toddlers Learn Language." *Child Development* 85, no. 3: 956–970.

Rowe, M. L., and A. Weisleder. 2020. "Language Development in Context." *Annual Review of Developmental Psychology* 2: 201–223.

Ruckenstein, M. 2023. *The Feel of Algorithms*. University of California Press.

Sandvig, C., K. Hamilton, K. Karahalios, and C. Langbort. 2014. "Auditing Algorithms: Research Methods for Detecting Discrimination on Internet Platforms." In *Data and Discrimination: Converting Critical Concerns into Productive Inquiry*, 1–23. ICA.

Seaver, N. 2022. *Computing Taste: Algorithms and the Makers of Music Recommendation*. University of Chicago Press.

Shamatov, D., and S. Bahry. 2020. "Variation in Educational Quality in Kyrgyzstan by District and Language of Instruction: An Analysis of 2017 National Scholarship Test Results." In *Education in Central Asia: A Kaleidoscope of Challenges and Opportunities*, 21–35. Springer International Publishing.

Shermatova, F., M. Totobayev, and S. Bahry. 2025. "Language Beliefs, Attitudes, and Practices in Contemporary Kyrgyzstan With Particular Reference to English." *International Journal of the Sociology of Language* 2025, no. 294: 99–123.

Sherouse, P. 2020. "Channels of Human Communication." In *The International Encyclopedia of Linguistic Anthropology*, edited by James Stanlaw, 1–9. Wiley.

Simon, E., R. Lima Jr., and L. De Cuypere. 2016. "Acquiring Non-Native Speech through Early Media Exposure: Belgian Children's Productions of English Vowels." *Poznan Studies in Contemporary Linguistics* 52, no. 4: 719–743.

Smagulova, J. 2021. "When Language Policy Is Not Enough." *International Journal of the Sociology of Language* 2021, no. 267–68: 265–269.

Smith, B., and A. Barad. 2018. "Digital Gaming and the Arts of Parental Control in Southern Peru: Phatic Functionality and Networks of Socialization in Processes of Language Socialization." *Journal of Linguistic Anthropology* 28, no. 3: 384–399.

Smith, J. 2006. "Non-Russians in the Soviet Union and After." In *The Cambridge History of Russia*, edited by M. Perrie, 495–521. Cambridge University Press.

Spitulnik, D. 1996. "The Social Circulation of Media Discourse and the Mediation of Communities." *Journal of Linguistic Anthropology* 6, no. 2: 161–187.

Sultanalieva, S. 2023. "Western and Post-Soviet Feminisms: Main Topics and Debates." In *"Nomadity of Being" in Central Asia: Narratives of Kyrgyzstani Women's Rights Activists*, 17–30. Springer Nature Singapore.

Sundqvist, P. 2009. "Extramural English Matters: Out-of-School English and Its Impact on Swedish Ninth Graders' Oral Proficiency and Vocabulary." PhD diss, Karlstad University.

Tomasello, M. 2010. *Origins of Human Communication*. MIT Press.

Umetbaeva, D. 2015. "Paradoxes of Hegemonic Discourse in Post-Soviet Kyrgyzstan: History Textbooks' and History Teachers' Attitudes Toward the Soviet Past." *Central Asian Affairs* 2, no. 3: 287–306.

UNESCO Ad Hoc Expert Group on Endangered Languages. 2003. "Language Vitality and Endangerment." In *Proceedings of the Document Submitted to the International Expert Meeting on UNESCO Programme Safeguarding of Endangered Languages, Paris, March 10–12, 2003. Paris*. UNESCO.

World Bank. 2025. "Personal Remittances, Received (% of GDP)—Kyrgyz Republic." World Bank Open Data. Accessed April 5, 2025.

Wu, S., M.-A. Rizoiu, and L. Xie. 2019. "Estimating Attention Flow in Online Video Networks." *Proceedings of the ACM on Human-Computer Interaction* 3, no. CSCW: Article 183, 1–25.

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